



I2C - BMA280 테스트

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Mango-M32F2 I2C-BMA280 3-Axis sensor 테스트

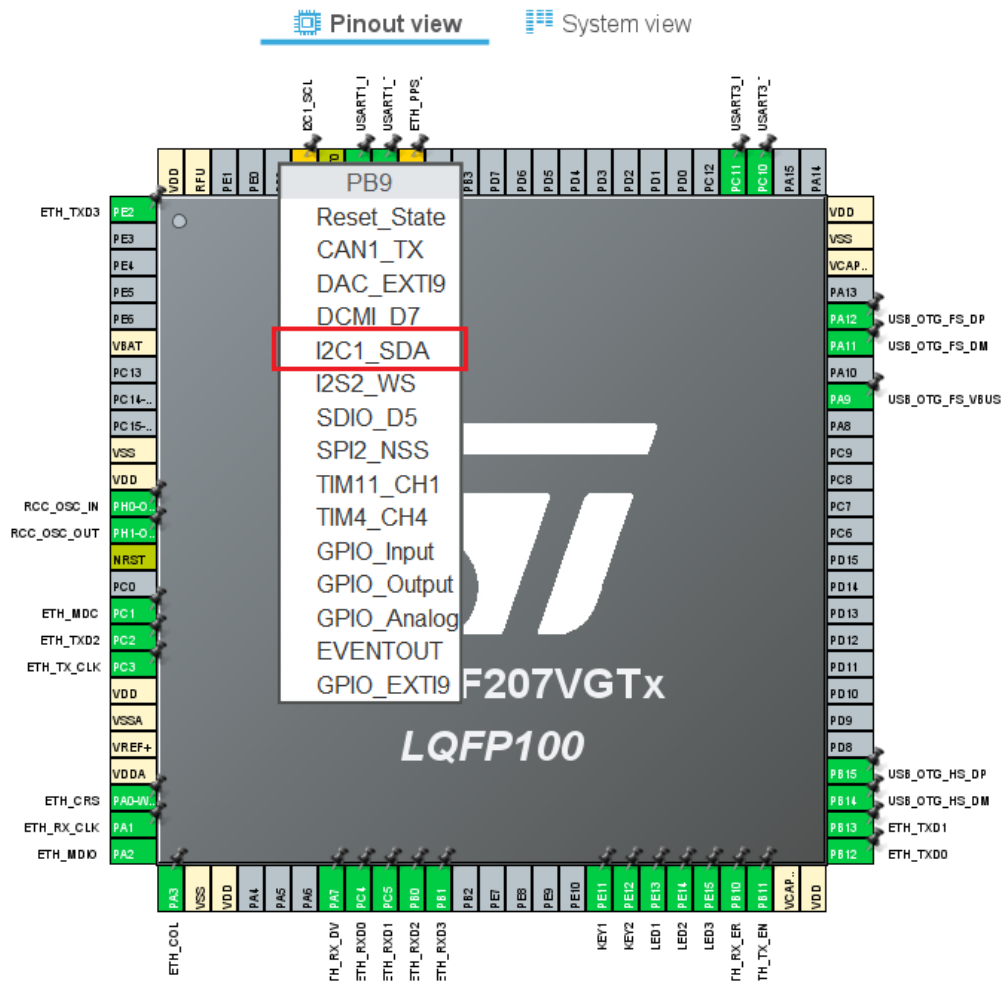
I2C pinmap

I2C – PB8, PB9

PB7	93	PB7	UART1_TX
PB8	95	PB8	I2C1_SCL
PB9	96	PB9	I2C1_SDA
PB10	47	PB10	MII_RX_FR

Pinout view

■ PB8-I2C1_SCL, PB9-I2C1_SDA 선택.



I2C 설정

- “Connectivity” 클릭, “I2C1” 클릭.
- “Mode” - “I2C” 클릭.

The screenshot displays the 'bma280_prj.ioc - Pinout & Configuration' window. The 'Pinout & Configuration' tab is active, showing a list of components on the left and configuration options on the right. In the left sidebar, under the 'Connectivity' category, 'I2C1' is selected and highlighted with a red box. The main panel shows the 'I2C1 Mode and Configuration' settings. The 'Mode' dropdown menu is set to 'I2C' and is also highlighted with a red box. Below this, the 'Configuration' section includes a 'Reset Configuration' button and tabs for 'NVIC Settings', 'DMA Settings', 'GPIO Settings', 'Parameter Settings' (which is selected), and 'User Constants'. The 'Parameter Settings' tab shows the following configuration parameters:

Configure the below parameters :	
Search (Ctrl+F)	
Master Features	
I2C Speed Mode	Standard Mode
I2C Clock Speed (Hz)	100000
Slave Features	
Clock No Stretch Mode	Disabled
Primary Address Length	7 bit

BMA280 Test Project

■ Code 생성 후 main.c 수정

```
36 /* Private define ----- */
37 /* USER CODE BEGIN PD */
38
39 #define BMA280_I2C_ADDR ((0x18 << 1)|
40
41 #define BMA280_RA_CHIP_ID          0x00
42 #define BMA280_RA_X_AXIS_LSB      0x02
43 #define BMA280_RA_BANDWIDTH        0x10
44 #define BMA280_RA_RANGE_BWIDTH     0x0f
45
46 /* range and bandwidth */
47 #define BMA280_RANGE_2G             0x3
48 #define BMA280_RANGE_4G             0x5
49 #define BMA280_RANGE_8G             0x8
50
51 #define BMA280_BW_7HZ                0
52 #define BMA280_BW_7_81HZ             0x8
53 #define BMA280_BW_15_13HZ            0x9
54 #define BMA280_BW_125HZ              12
55 #define BMA280_BW_250HZ              13
56 #define BMA280_BW_500HZ              14
57 #define BMA280_BW_MAX                0xF
58
59 /* USER CODE END PD */
```

BMA280 Test Project

```
89  /* USER CODE BEGIN 0 */
90  static unsigned int count = 0;
91  |
92  static uint8_t Read_1_Byte(uint8_t address)
93  {
94      uint8_t Buffer_Tx;
95      uint8_t Buffer_Rx;
96
97      Buffer_Tx = address;
98
99      HAL_I2C_Master_Transmit(&hi2c1, BMA280_I2C_ADDR, &Buffer_Tx, 1, 1000);
100     HAL_I2C_Master_Receive(&hi2c1, BMA280_I2C_ADDR, &Buffer_Rx, 1, 1000);
101
102     return Buffer_Rx;
103 }
104
105 static uint8_t Get_ChipID(void)
106 {
107     return Read_1_Byte(BMA280_RA_CHIP_ID);
108 }
109
110 /*
111  * Description : none.
112  * Argument(s) : none.
113  * Return(s)   : none.
114  * Note(s)     : none.
115  */
116 static void Read_3_Axis_BMA280_Data(int16_t * xData_p, int16_t * yData_p, int16_t * zData_p)
117 {
118     uint8_t Buffer_Tx;
119     uint8_t Buffer_Rx[6];
120
121     Buffer_Tx = BMA280_RA_X_AXIS_LSB;
122
123     HAL_I2C_Master_Transmit(&hi2c1, BMA280_I2C_ADDR, &Buffer_Tx, 1, 1000);
124     HAL_I2C_Master_Receive(&hi2c1, BMA280_I2C_ADDR, Buffer_Rx, 6, 1000);
125
126     * xData_p = (((int16_t)(int8_t)Buffer_Rx[1]) << 8) | Buffer_Rx[0] >> 6;
127     * yData_p = (((int16_t)(int8_t)Buffer_Rx[3]) << 8) | Buffer_Rx[2] >> 6;
128     * zData_p = (((int16_t)(int8_t)Buffer_Rx[5]) << 8) | Buffer_Rx[4] >> 6;
129 }
130 /* USER CODE END 0 */
```

BMA280 Test Project

```
196  MX_I2C1_Init();
197  /* USER CODE BEGIN 2 */
198  printf("I2C-BMA280 test\r\n");
199  httpd_init();
200
201  printf("Get_ChipID: %X\n", Get_ChipID());
202
203  Buffer_Tx[0] = BMA280_RA_RANGE_BWIDTH;
204  Buffer_Tx[1] = BMA280_RANGE_8G << 3;
205  HAL_I2C_Master_Transmit(&hi2c1, BMA280_I2C_ADDR, Buffer_Tx, 2, 1000);
206  Buffer_Tx[0] = BMA280_RA_BANDWIDTH;
207  Buffer_Tx[1] = BMA280_BW_500HZ;
208  HAL_I2C_Master_Transmit(&hi2c1, BMA280_I2C_ADDR, Buffer_Tx, 2, 1000);
209  printf("Set Param BMA280_RANGE_8G, BMA280_BW_500HZ\n");
210  /* USER CODE END 2 */
211
212  /* Infinite loop */
213  /* USER CODE BEGIN WHILE */
214  while (1)
215  {
216      MX_LWIP_Process();
217
218      if(count++ % 10000 == 0)
219      {
220          Read_3_Axis_BMA280_Data(&xData, &yData, &zData);
221          printf("x: %d, y: %d, z: %d\n", xData, yData, zData);
222      }
223  /* USER CODE END WHILE */
224  MX_USB_HOST_Process();
225
226  /* USER CODE BEGIN 3 */
227  }
228  /* USER CODE END 3 */
229 }
```


Test 결과 확인

- CR-zF-BMA280 센서 보드를 8Pin Cable을 이용 CON7에 연결함.
- Chip ID – 0xFB 정상적으로 읽힘.
- 센서를 움직일 때 3-axis 데이터가 변동함.

COM4 - Tera Term VT

File Edit Setup Control Window Help

```
I2C-BMA280 test
Get_ChipID: FB
Set Param BMA280_RANGE_8G, BMA280_BW_500HZ
x: 162, y: 180, z: -53
x: 163, y: 217, z: -282
x: 121, y: 247, z: -213
x: 26, y: 454, z: -344
x: 82, y: 421, z: 21
x: 511, y: 220, z: 511
x: -51, y: 156, z: 32
x: -186, y: 511, z: -319
x: -63, y: -118, z: 189
x: -16, y: 511, z: 27
x: 511, y: 472, z: 511
x: 349, y: 415, z: 148
x: -158, y: 48, z: 4
x: -172, y: 17, z: 25
x: 171, y: 237, z: 172
x: 272, y: 445, z: 511
x: 33, y: 258, z: 91
x: 323, y: 222, z: 173
x: 306, y: 264, z: 51
x: 238, y: 150, z: 44
```

Thank You

